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Learning the Basics of Wireshark and Network Traffic

In this lab, I learned how to use Wireshark to look at network traffic. The main goal was to understand how data moves across a network and how Wireshark can be used to capture and analyze that data. Wireshark is used to capture and analyze network traffic so you can see how data moves between devices and what protocols are being used. This lab helped build a basic foundation for understanding how computers communicate with each other.

Tools I Used

- Wireshark
- Kali Linux
- Web Browser

Skills I Learned

- Capturing live network traffic using Wireshark
- Opening and reviewing packet capture files
- Identifying IP addresses and port numbers
- Recognizing common network protocols
- Understanding how data moves across a network

Task 1) setup

1. Logged into the Cyber Range system and started the Kali Linux virtual machine.
2. Logged into the Kali Linux desktop using the provided login information.
3. Opened Wireshark from the applications menu.
4. Viewed the main Wireshark screen, including the packet list, packet details, and packet bytes sections.
5. Identified the available network interfaces shown on the start screen.

This section helped me understand how Wireshark is laid out and where information is displayed. Knowing the interface makes it easier to analyze packets later in the lab.

Task 2) Capturing Live Network Traffic

1. Selected the active network interface in Wireshark.
2. Clicked Start Capture to begin capturing packets.
3. Opened a web browser.
4. Typed a website URL into the browser to generate network traffic.
5. Returned to Wireshark and observed packets appearing in real time.
6. Clicked Stop Capture after enough traffic was collected.

This part showed how Wireshark captures data in real time. Visiting a website made it clear how network traffic is created and recorded.

Task 3) Opening a Saved Packet Capture

1. Downloaded the provided packet capture (.pcap) file.
2. Opened Wireshark.
3. Clicked File then Open.
4. Selected the downloaded packet capture file.
5. Viewed the list of packets contained in the file.
6. Clicked different packets to view their details.

A packet capture shows source and destination IP addresses, port numbers, protocols, and packet data, which helps explain how systems communicate. Opening a saved packet capture made it possible to analyze traffic without capturing it live. This is useful when reviewing past activity.

Task 4) Viewing IP Addresses and Port Numbers

1. Selected a packet from the packet list.
2. Viewed the source IP address and destination IP address in the packet details.
3. Identified the source port and destination port numbers.
4. Repeated this process with multiple packets.
5. Noted how different services use different ports.

This section showed how to tell where data is coming from and where it is going. IP addresses and ports help identify communication between systems.

Task 5) Identifying Common Network Protocols

1. Looked at the Protocol column in Wireshark.
2. Identified packets using TCP and UDP.
3. Found DNS packets and reviewed their details.
4. Found DHCP packets related to IP address assignment.
5. Compared how different protocols appear in Wireshark.

Packet analysis helps troubleshoot network problems and understand normal versus unusual traffic. It can also help identify security issues. This section helped me recognize different types of network traffic. Understanding protocols makes it easier to understand how networks work.